Chapter 3: Strategic Interaction and Basic Game Theory

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Outline

1. Introduction and Motivation
2. Basic Concepts in Game Theory
3. Nash Equilibrium
4. Extensive-Form Games (Sequential)
5. Applications and Examples
6. Exercises and Wrap-Up

Why Study Game Theory?

- **Beyond Price-Taking**: In many markets, agents (firms, individuals) make decisions *strategically*, anticipating others' responses.
- **Micro Foundations**: Oligopoly pricing, auctions, bargaining, voting—these all involve strategic interaction.
- **Policy Relevance**: Understanding strategic behavior is essential for regulation, antitrust, and mechanism design.

Key Goal

Formalize **strategic interaction** using game theory concepts like *Nash equilibrium*, *dominant strategies*, and more.

Roadmap for Chapter 3

1. **Basic Concepts**
   - Normal-Form Games
   - Dominant/Dominated Strategies
   - Nash Equilibrium (Pure vs. Mixed)
2. **Extensive-Form Games (Sequential)**
   - Game Trees, Perfect Information
   - Subgame-Perfect Equilibrium
3. **Applications**
   - Classic Games (PD, Coordination)
   - Oligopoly Revisited
   - Simple Bargaining, Repeated Games
4. **Exercises & Extensions**

What is a Game?

Definition (Informal): A situation in which multiple agents (players)

- have **objectives** (utilities, payoffs),
- make **choices** (strategies),
- and the outcome (payoff) depends on **all** players' choices.

Strategy

A *complete plan* of actions a player will take, given any possible situation they face in the game.

Normal-Form (Strategic Form) Game

A normal-form game is given by:

$$\Big\langle \text{Players, (Strategy Sets), (Payoff Functions)} \Big\rangle.$$

- **Players**: Usually labeled $1, 2, \ldots, n$.
- **Strategy Sets**: For each player i, strategy set S_i.
- **Payoff Functions**: $u_i(s_1, s_2, \ldots, s_n)$ for each player, depending on the strategy profile.

Example (Example)

A 2-player game can be represented by a payoff matrix. Each cell shows (u_1, u_2) for the chosen strategies (s_1, s_2).

Example: A Simple 2x2 Game

	L	R
U	$(2, 2)$	$(0, 3)$
D	$(3, 0)$	$(1, 1)$

- **Players**: 1 (chooses rows) and 2 (chooses columns).
- **Strategies**: Player 1 has $\{U, D\}$; Player 2 has $\{L, R\}$.
- **Payoffs**: For each cell, the tuple is (u_1, u_2).

Question

Which outcome do we expect if both players are rational and strategic?

Dominant and Dominated Strategies

Strictly Dominant Strategy:

- Strategy s_i^* is **strictly dominant** for player i if it yields a strictly higher payoff than all other strategies, *regardless of what others do*:

$$u_i(s_i^*, s_{-i}) > u_i(s_i', s_{-i}), \quad \forall s_{-i}, \quad \forall s_i' \neq s_i^*.$$

Strictly Dominated Strategy:

- Strategy s_i' is **strictly dominated** if there exists another strategy s_i^* such that:

$$u_i(s_i^*, s_{-i}) > u_i(s_i', s_{-i}), \quad \forall s_{-i}.$$

Remark

Dominant strategies are not always present; dominated strategies are often easier to spot and eliminate.

Iterated Elimination of Dominated Strategies (IEDS)

- Start by **removing** any strictly dominated strategies for each player.
- This reduction might make new strategies become dominated.
- Keep iterating until no more dominated strategies can be eliminated.
- The resulting "reduced game" may have fewer strategy profiles, simplifying the search for equilibrium.

IEDS Example

Example (An Illustrative Example)

Payoff matrix *(2 players, 3 strategies each):*

Player 1 \ *Player 2*	A_2	B_2	C_2
A_1	$(2,3)$	$(3,4)$	$(1,2)$
B_1	$(3,1)$	$(4,1)$	$(2,3)$
C_1	$(2,4)$	$(5,2)$	$(1,4)$

- *First, notice for Player 2: C_2 might dominate B_2 after Player 1's A_1 is removed (or vice versa).*
- *Once we remove the dominated strategy for Player 1, we re-check Player 2's payoffs, etc.*

This iterative removal can reveal new dominated strategies at each step.

Nash Equilibrium (NE)

Idea: Each player's strategy is a **best response** to the other players' strategies.

Definition (Nash Equilibrium)

A strategy profile $(s_1^*, \ldots, s_n^*)$ is a *Nash equilibrium* if for every player i,

$$s_i^* \in \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}^*).$$

i.e. no player can *unilaterally* deviate and gain a higher payoff.

Key Point

No player has an incentive to deviate **once** we are at $(s_1^*, \ldots, s_n^*)$.

Classic Example: Prisoner's Dilemma (PD)

- Two suspects. If both remain silent, each gets a light sentence.
- If one testifies (C) while the other is silent (S), testifier goes free while the other gets heavy penalty.

	Confess (C)	Silent (S)
Confess (C)	$(-5, -5)$	$(0, -10)$
Silent (S)	$(-10, 0)$	$(-1, -1)$

Observations

- Both players have a **strictly dominant strategy** to confess (C). - (C, C) is a **Nash equilibrium** but yields worse payoff than (S, S) for both.

Example: Coordination Game

	A	B
A	$(2,2)$	$(0,0)$
B	$(0,0)$	$(1,1)$

- (A, A) and (B, B) are both **NE**.
- No dominant strategy; one's best response depends on what the other does.

Remark

We have **multiple equilibria**. *This can lead to coordination failure if each anticipates the other picking a different action.*

Existence of Nash Equilibrium (in Mixed Strategies)

- **Pure strategy NE** might not exist (e.g. matching pennies).
- **Mixed strategy NE**: players randomize strategies with certain probabilities to make the opponent indifferent.
- **Key Theorem** (Nash, 1950s): Every finite game has at least one NE (possibly in mixed strategies).

Interpretation

Even if a pure strategy equilibrium doesn't exist, rational players can resort to probability distributions over actions that **mutually best-respond**.

Example: Matching Pennies

- Two players simultaneously choose Heads or Tails.
- If same side: Player 1 wins (+1), Player 2 loses (-1). If different sides: Player 1 loses, Player 2 wins.

	Heads	Tails
Heads	$(+1, -1)$	$(-1, +1)$
Tails	$(-1, +1)$	$(+1, -1)$

No Pure NE

One can see no pure strategy is stable. But a **mixed strategy** NE exists: each player randomizes $50\% - 50\%$.

Computing a Mixed-Strategy NE (Example)

General Approach:

1. Suppose each player randomizes among actions so as to **make the other player indifferent** among *their* possible actions.
2. Solve these indifference conditions simultaneously to find equilibrium probabilities.

Concrete Example: Rock-Paper-Scissors (brief):

- Each player chooses R, P, or S. Payoffs: 0 if tie; $+1$ for a win, -1 for a loss.
- If you believe the opponent plays R with prob x, P with prob y, S with prob $1 - x - y$, you pick your strategy to maximize expected payoff.
- **Symmetry** suggests $x = y = 1/3$ in equilibrium, each is indifferent across R, P, S.

Key Idea

In a mixed NE, each player's *mix* must make the other *indifferent* across the strategies they are mixing over.

Summary: Nash Equilibrium

- **Definition**: No unilateral deviation profitable.
- **Pure vs. Mixed**: Some games have pure NE, some only have mixed NE, and some have both.
- **Multiplicity**: Coordination games can have multiple NE.
- **Efficiency**: NE need not be socially optimal (e.g. Prisoner's Dilemma).

Extensive Form: Sequential Move Games

- In many economic settings, players move **in sequence** (e.g., Stackelberg duopoly).
- Represented by a **game tree**: each node indicates whose turn it is, possible actions, and payoffs at terminal nodes.
- We often assume **perfect information**: each player knows exactly the history of moves made so far when choosing an action (no simultaneous moves, no hidden info).

Note on Terminology

- **Perfect information** = each player sees all prior actions. - **Complete information** = all payoffs/strategy sets are known to all players, but not necessarily the order of moves or the moves themselves. They are different concepts.

Backward Induction: Logic

1. Start from the **last mover**: determine best action(s) at each possible node.
2. Move one step earlier: anticipating the last mover's best response, pick the best action for the second-to-last mover.
3. Continue until reaching the initial node.

Example (Stackelberg Duopoly Recall)

- **Stage 2**: *Follower chooses q_F best response to q_L.*
- **Stage 1**: *Leader anticipates q_F^*, picks q_L to maximize profit.*
- *This yields a different outcome than Cournot (simultaneous).*

Subgame-Perfect Equilibrium (SPE)

Definition (SPE)

A strategy profile is **subgame-perfect** if it constitutes a Nash equilibrium *in every subgame* of the original game.

- Eliminates **non-credible threats**: A threat a rational player wouldn't carry out in a later subgame is not valid in SPE.
- Found via **backward induction** in finite **perfect-information** games.

SPE Example

Example (Simple Example: Ultimatum Game)

- *Player 1 proposes how to split \$10. Player 2 accepts or rejects.*
- **Nash Equilibrium**: *Player 1 proposes (\$5, \$5), and Player 2 accepts any offer of \$5 or more.*
- **Lack of Credibility**: *We see subgame perfection clarifies that if Player 2 is rational and cares only about money, they accept any* > 0*; so the equilibrium is* $(\$9.99, \$0.01)$ *in the limit.*

A subgame-perfect equilibrium cannot involve non-credible threats (e.g. "I will reject anything less than \$5").

Application: Oligopoly Revisited

- **Cournot Model**: Simultaneous choice of quantities $\rightarrow$ normal-form representation, NE solution.
- **Bertrand Model**: Simultaneous price choice $\rightarrow$ also a normal-form game, NE is $p^* = MC$ if identical products/costs.
- **Stackelberg Model**: Sequential quantity choice $\rightarrow$ extensive-form representation, *SPE* found by backward induction.

Link to Chapter 2

These models differ in move order (simultaneous vs. sequential) and strategic variable (price vs. quantity), but all are game-theoretic.

Bargaining: A Simple Example

Two players, A and B, want to split \$100.

- **Stage 1**: A proposes how to divide the \$100, e.g. $(x, 100 - x)$.
- **Stage 2**: B either accepts or rejects.
- If B accepts, the split happens as proposed. If B rejects, both get 0 (or the game ends with no agreement).

Analysis via Backward Induction

- **Last mover** is B who decides Accept/Reject. If rejecting yields 0 for B, then B accepts *any* proposal that gives them > 0. - Anticipating this, A offers $(\$100 - \epsilon, \epsilon)$, with ϵ very small. **SPE** outcome is near $(\$100, \$0)$ favoring the proposer.

Repeated Games: A Simple Example

Repeated Prisoner's Dilemma:

- Stage game (one-shot PD) payoff matrix is known; unique NE is (C, C) (Confess, Confess).
- Suppose it is repeated *infinitely* with discount factor $\delta < 1$.

Trigger Strategy (Grim Trigger):

- Start cooperating (play S) unless the other side has ever defected (played C).
- If the other defects once, punish by playing C forever.

Folk Theorem (Intuitive)

If δ is large enough (players sufficiently patient), the threat of eternal punishment can sustain **mutual cooperation** in every period as a *subgame-perfect equilibrium*—an outcome that is better for both than repeated confession.

Policy & Mechanism Design Perspective

- **Policy Goal**: Often we want to incentivize certain behavior (cooperation, truth-telling, etc.).
- **Game Structure**: By changing rules (taxes, subsidies, legal constraints), we alter the *payoff functions* and thus the equilibria.
- **Mechanism Design**: Engineer the game so that the **desired outcome** becomes a (unique) equilibrium.

Example

Auction design: Setting payment rules and bidding formats to ensure efficiency or revenue. Public goods: VCG mechanisms to reveal true valuations.

Exercises (1)

1. **Dominant Strategy / IEDS**:
   - Given a 2-player game matrix, identify if any strategy is strictly dominant.
   - Apply Iterated Elimination of Dominated Strategies (IEDS).
   - Illustrate how removing one strategy can make another dominated.
2. **Nash Equilibrium in 2x2 Games**:
   - For a given payoff matrix, check all pure strategy profiles for NE. Are there multiple NE or none?

Exercises (2)

1. **Mixed Strategy NE**:
   - Construct a 2-player game with no pure NE (like Matching Pennies).
   - Solve for the mixed NE by making each player indifferent.
2. **Cournot Duopoly (Review)**:
   - Show how Cournot equilibrium is a **Nash equilibrium** in normal form.
   - If each firm has cost $C(q) = cq$, and inverse demand $p = a - Q$, find q_1^*, q_2^* explicitly.

Exercises (3)

1. **Stackelberg Extended**:
   - Two firms: leader chooses q_L, follower observes and chooses q_F.
   - Write the extensive form. Solve by backward induction for SPE (q_L^*, q_F^*). Compare to Cournot.
2. **Credible Threats / Subgame-Perfectness**:
   - Construct a small sequential game where a threat is not credible if the subgame is reached.
   - Show how backward induction or subgame perfection eliminates that threat.

Exercises (4)

1. **Repeated Prisoner's Dilemma**:
   - If it is repeated infinitely with discount factor δ, show how a *grim trigger* can sustain mutual cooperation if δ is large.
2. **Bargaining Model (2-Stage)**:
   - Two agents bargain over splitting \$100 in two rounds. If no agreement in Round 1, the pie shrinks to \$80 in Round 2.
   - Solve by backward induction for a subgame-perfect outcome.

References (1/2)

- Mas-Colell, Whinston, and Green. *Microeconomic Theory* (Game Theory chapters).
- Tirole, J. *The Theory of Industrial Organization* (Oligopoly, strategic behavior).
- Jehle & Reny. *Advanced Microeconomic Theory*.

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- Osborne, M. *An Introduction to Game Theory*.
- Fudenberg & Tirole. *Game Theory*.
- Binmore, K. *Playing for Real*.

Additional Notes

- Game theory underpins much of modern microeconomics, from oligopoly to mechanism design.
- We introduced the **basics**: normal form, extensive form, NE, subgame perfection.
- Future chapters will cover auctions, signaling games, incomplete information, and advanced mechanism design.

Key Takeaway

Strategic thinking is essential when agents anticipate each other's actions—beyond simple price-taking.

Concluding Remarks

- **Nash equilibrium** captures mutual best responses.
- **Subgame-perfect equilibrium** refines NE for sequential moves, removing non-credible threats.
- **Applications**: Oligopoly, auctions, bargaining, repeated games, policy design.

Next Topics

We will delve deeper into **incomplete information** and **signaling/screening** in subsequent chapters.

Q & A

Any Questions?

Thank you for your attention!